Lesson 13 Demo 01

Creating Virtual Networking and Resolving Domain Names

Objective: To create Azure's virtual networking capabilities by creating a virtual network for Azure virtual machines, implementing network-based segmentation with static IP addresses, securing public endpoints, and enabling DNS name resolution within the virtual network

Tools Required: Azure account with administrator access

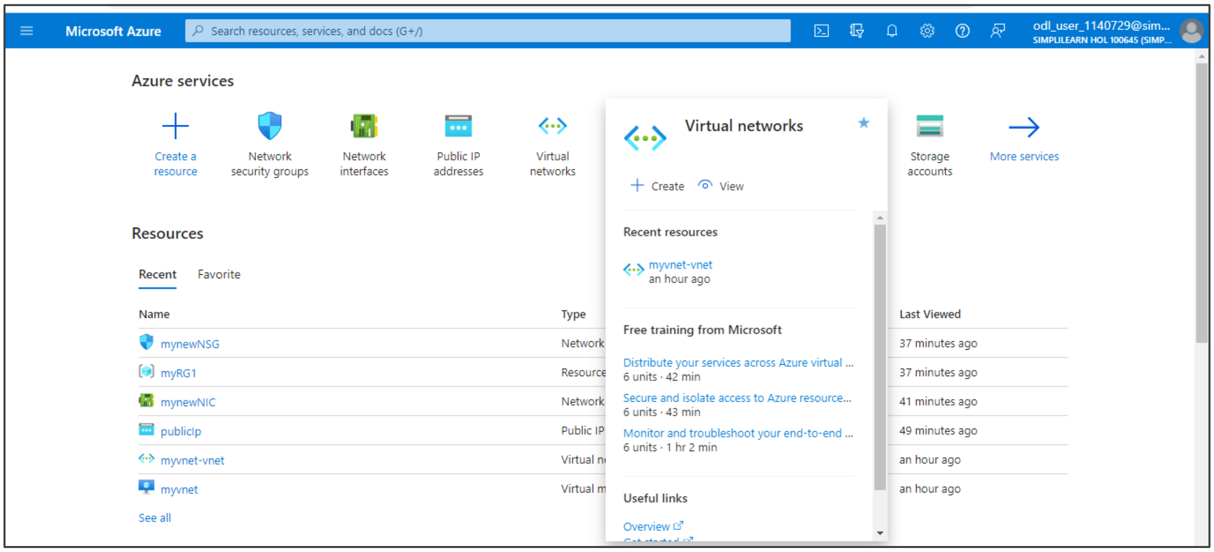
Prerequisites: None

Steps to be followed:

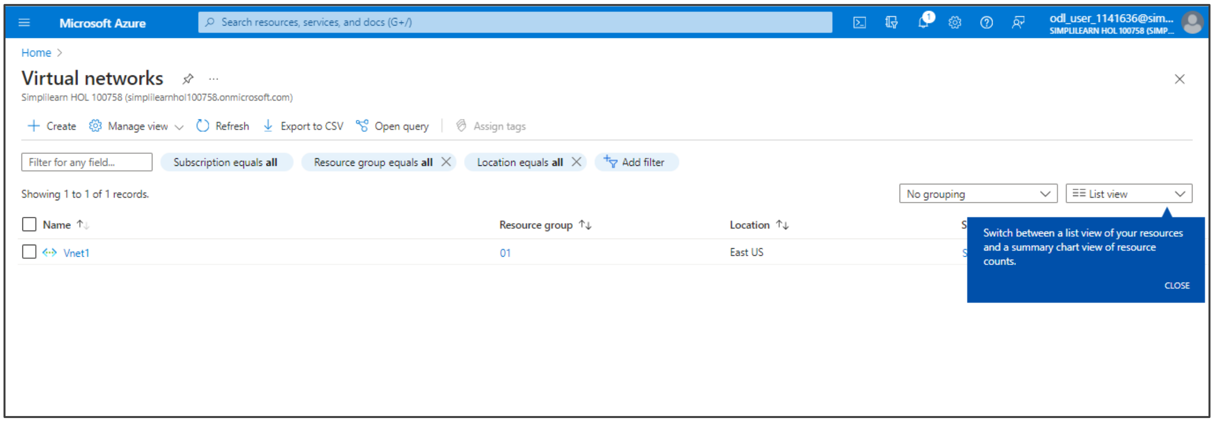
1. Create a virtual network
2. Create a subnet
3. Create a public IP address
4. Create the network interface
5. Create the network security group

Step 1: Create a virtual network

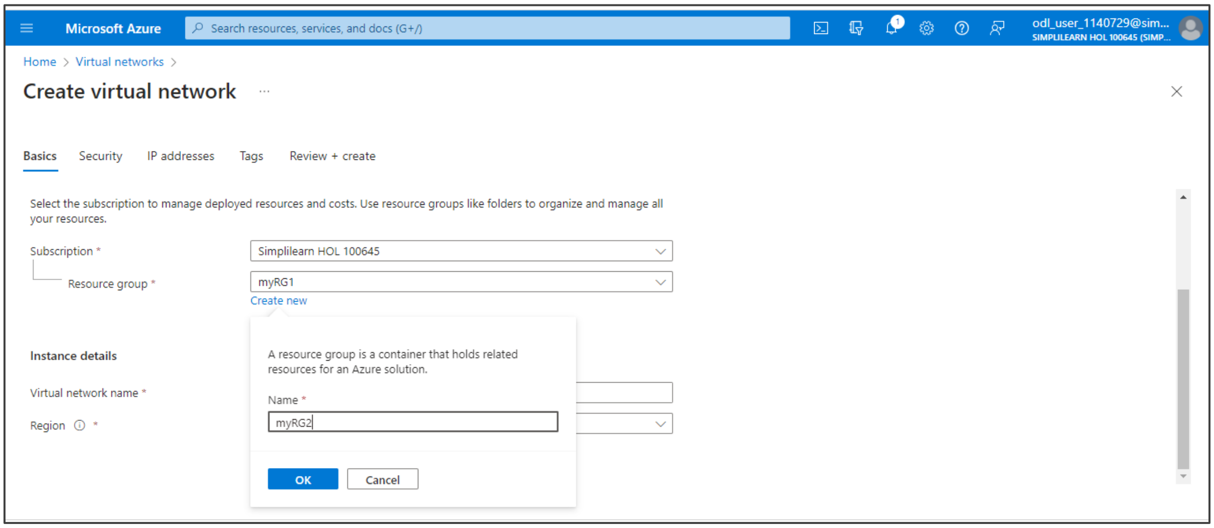
1.1 Log in to the Azure portal; search for and select **Virtual networks**



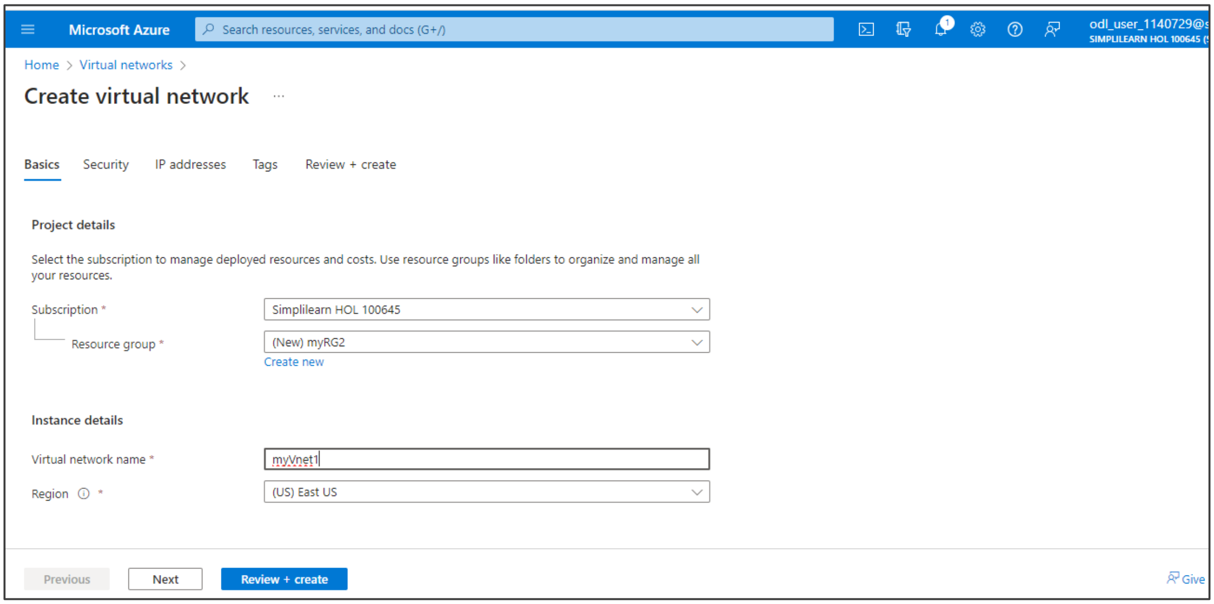
1.2 In the **Virtual networks** page, click on **+ Create**



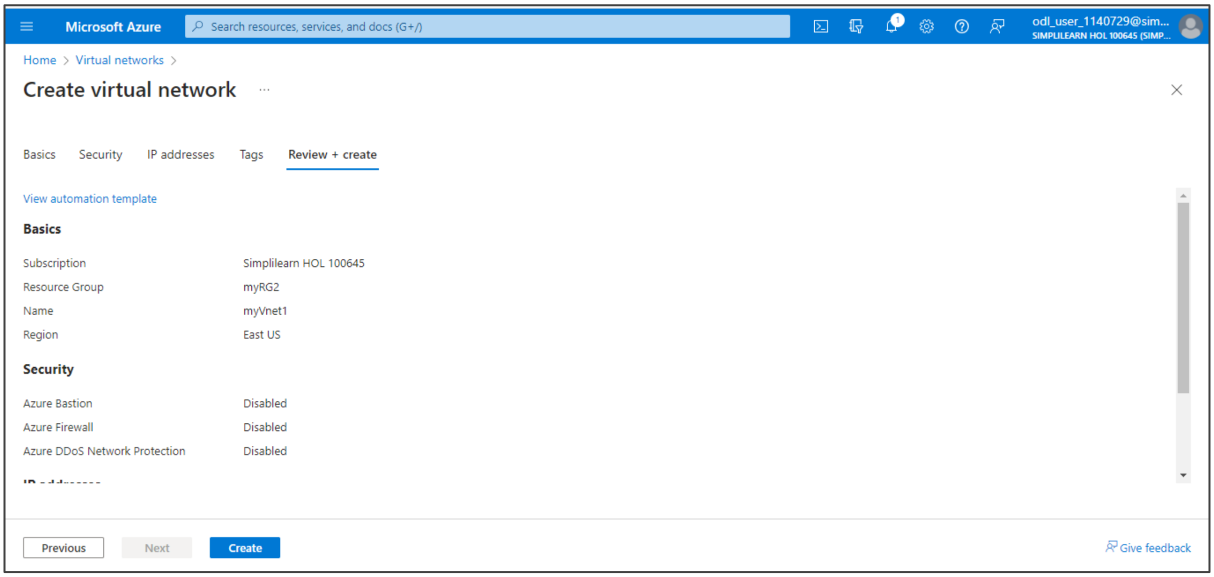
1.3 In the **Create virtual network** page, navigate to the **Basics** tab and under **Resource group**, click on **Create new** to create a new resource group (**myRG2**)



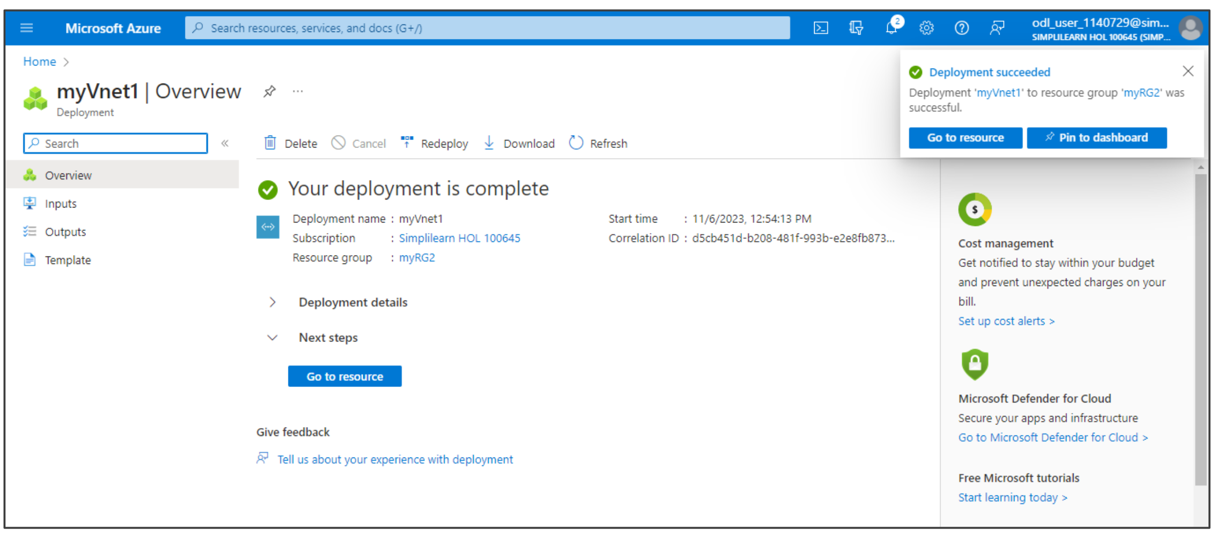
1.4 Under **Instance details**, enter the **Virtual network name** as **myVnet1** and click on **Review + create**



1.5 Click on **Create**

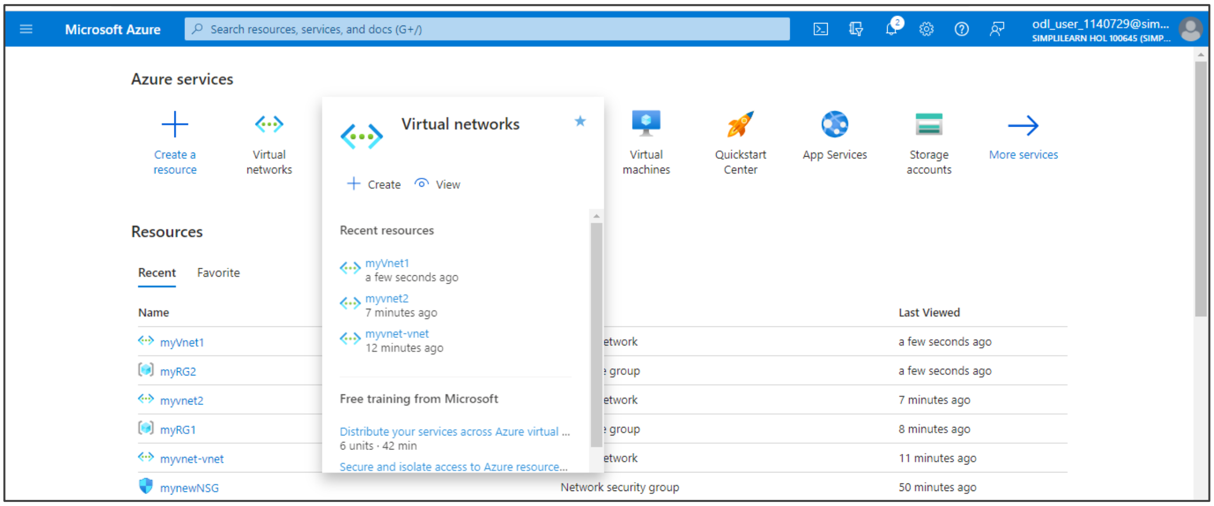


The output appears as shown below:

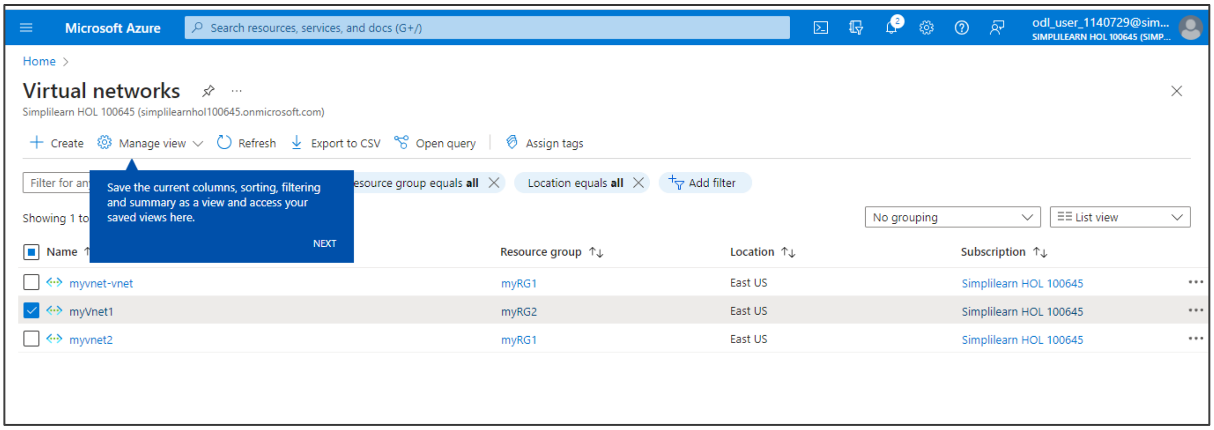


Step 2: Create a subnet

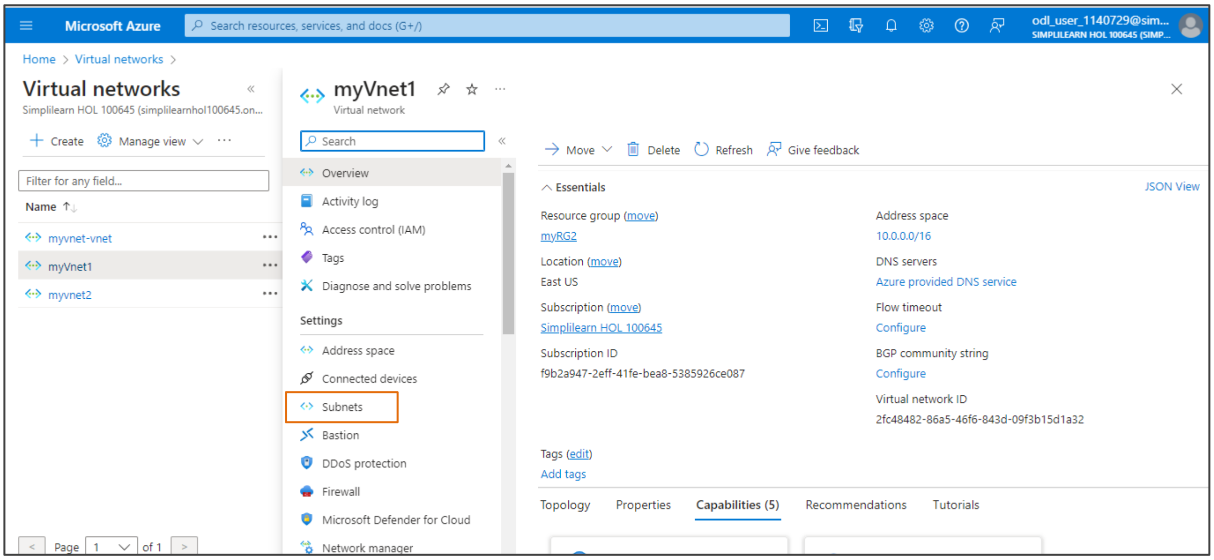
2.1 Navigate to **Virtual networks**



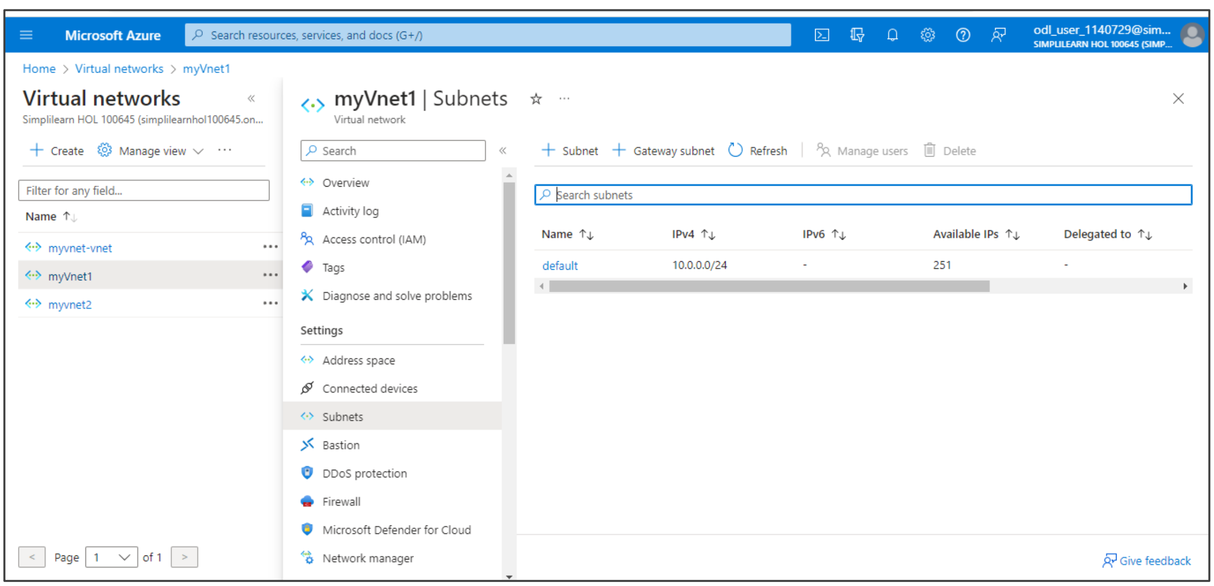
2.2 Choose the **Virtual network (myVnet1)** to which the user wants to add the subnet



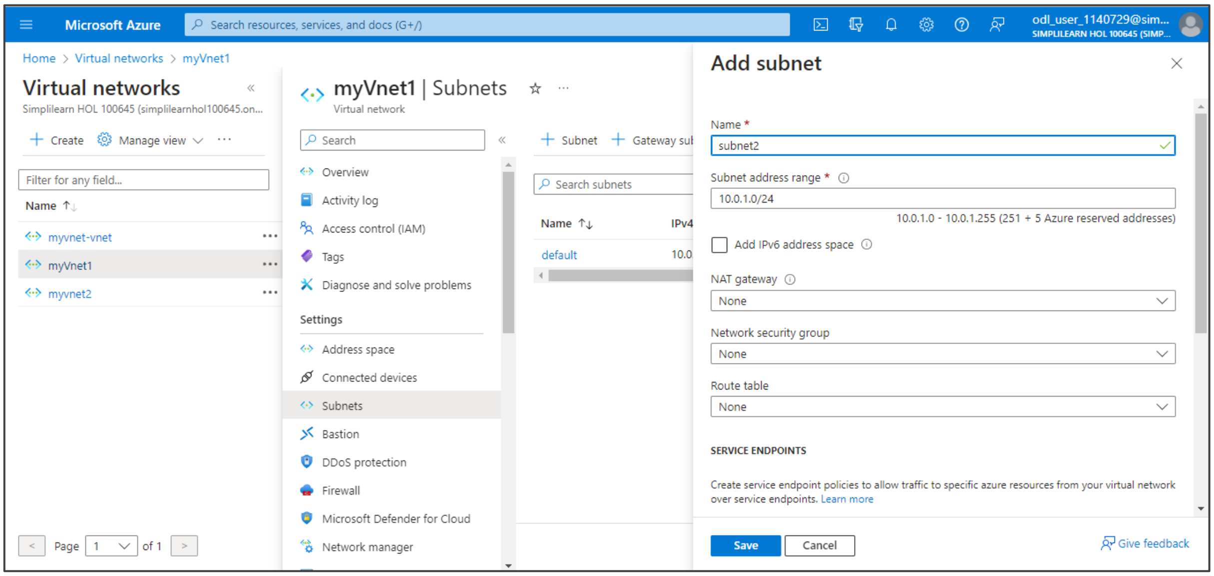
2.3 Under **Settings**, select **Subnets**



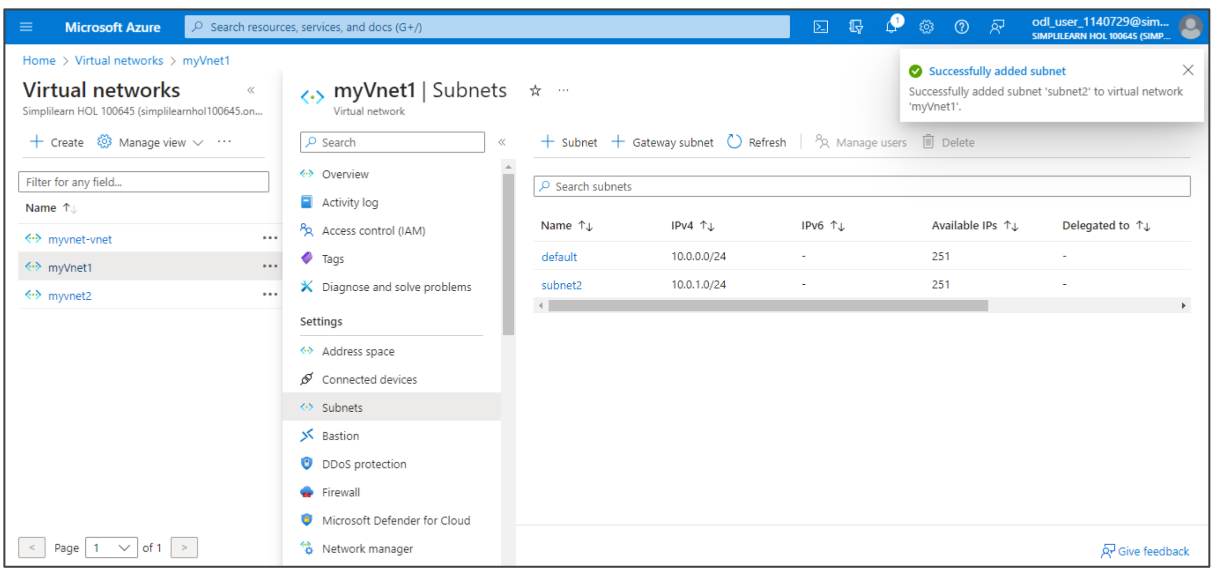
2.4 Click on **+ Subnet**



2.5 In the **Add subnet** page, enter the **Name** as **subnet2** and click on **Save**

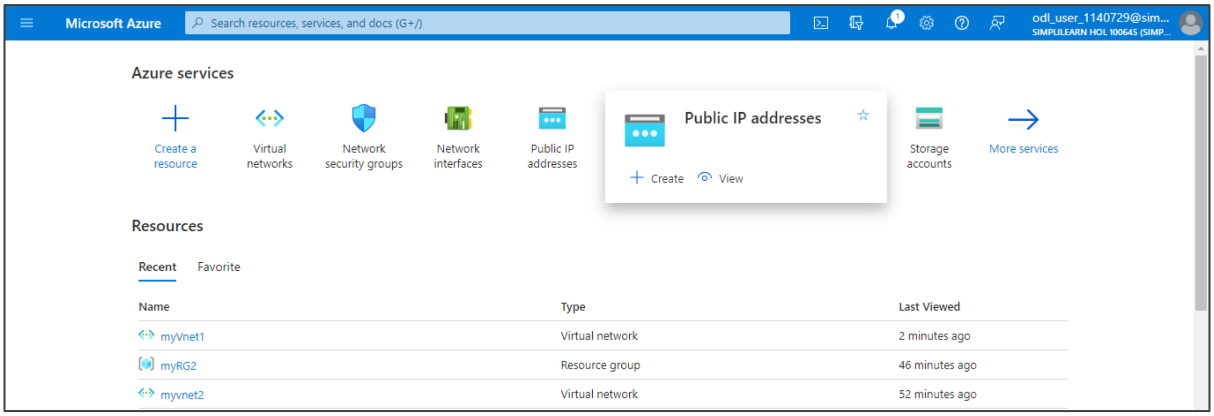


The output appears as shown below:

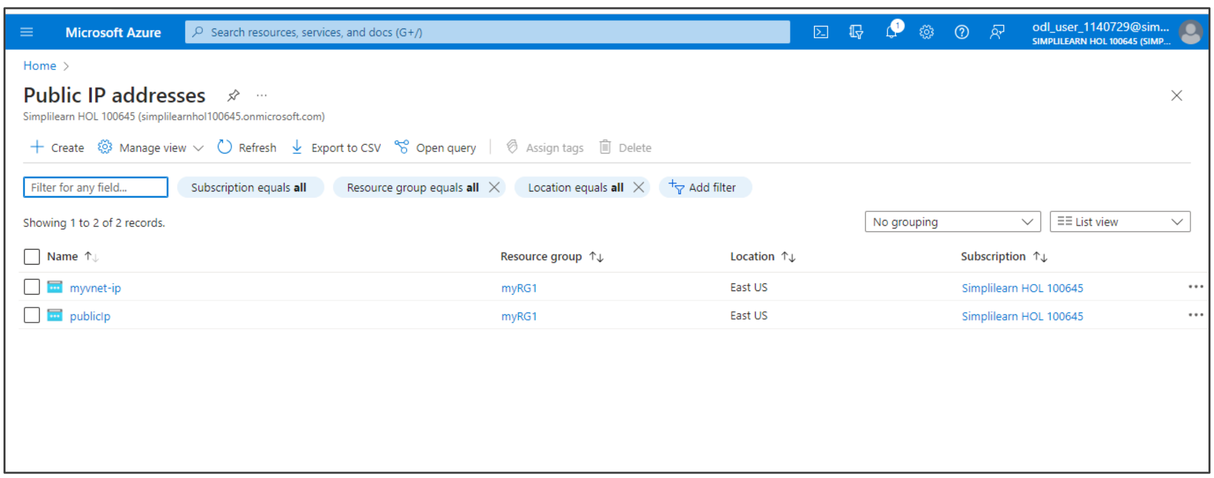


Step 3: Create a public IP address

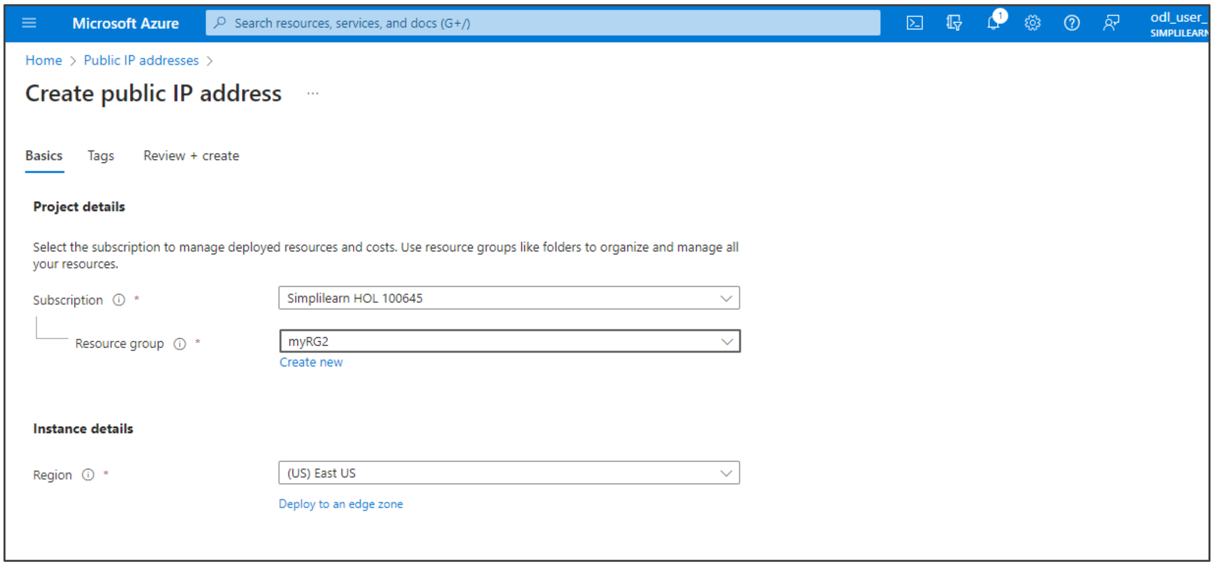
3.1 In the Azure portal, search for and select **Public IP addresses**



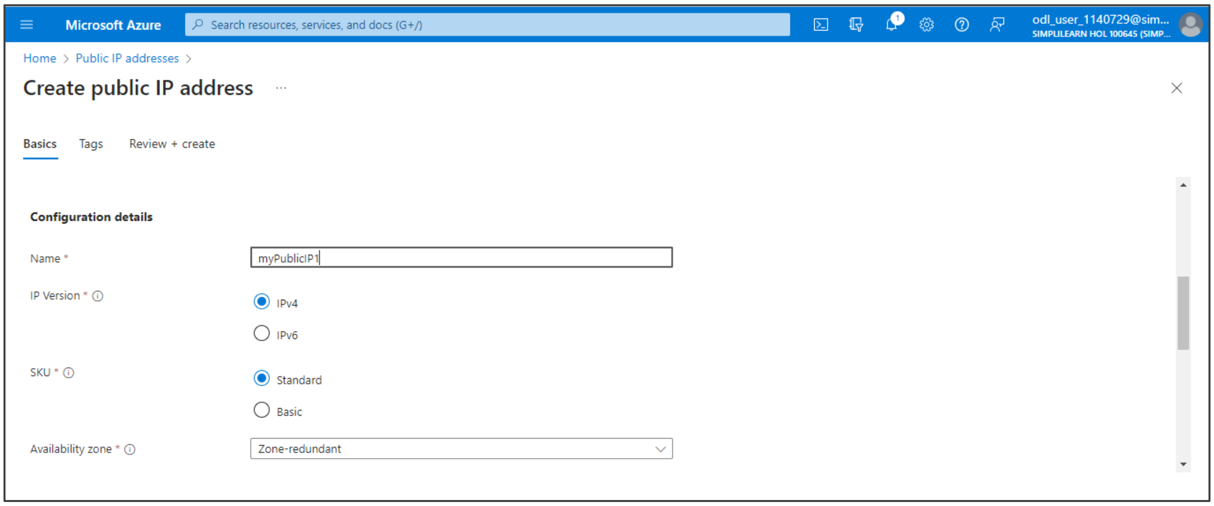
3.2 In the **Public IP addresses** page, click on **+ Create**

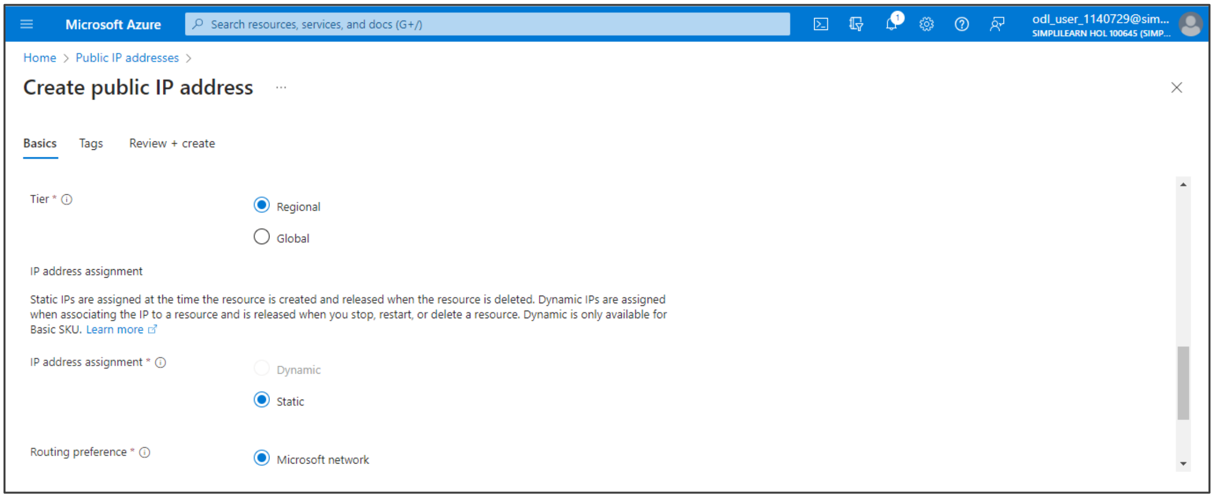


3.3 In the **Create public IP address** page, navigate to the **Basics** tab and enter the following details as shown in the screenshot below:

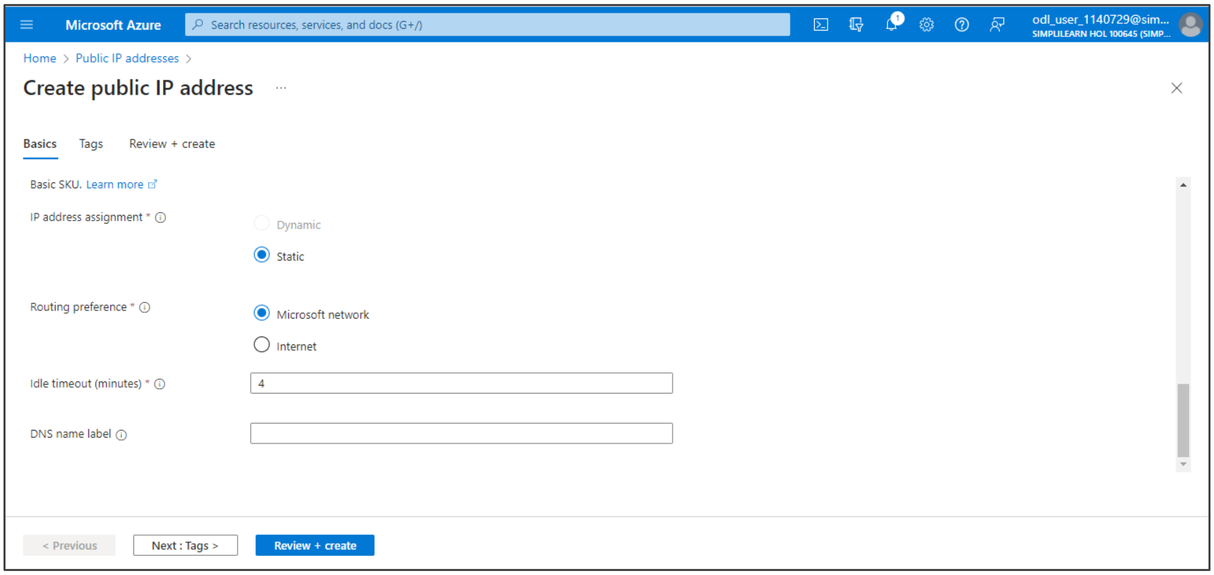


3.4 Under **Configuration details,** enter the **Name** as **myPublicIP1** and other details as shown in the screenshots below:

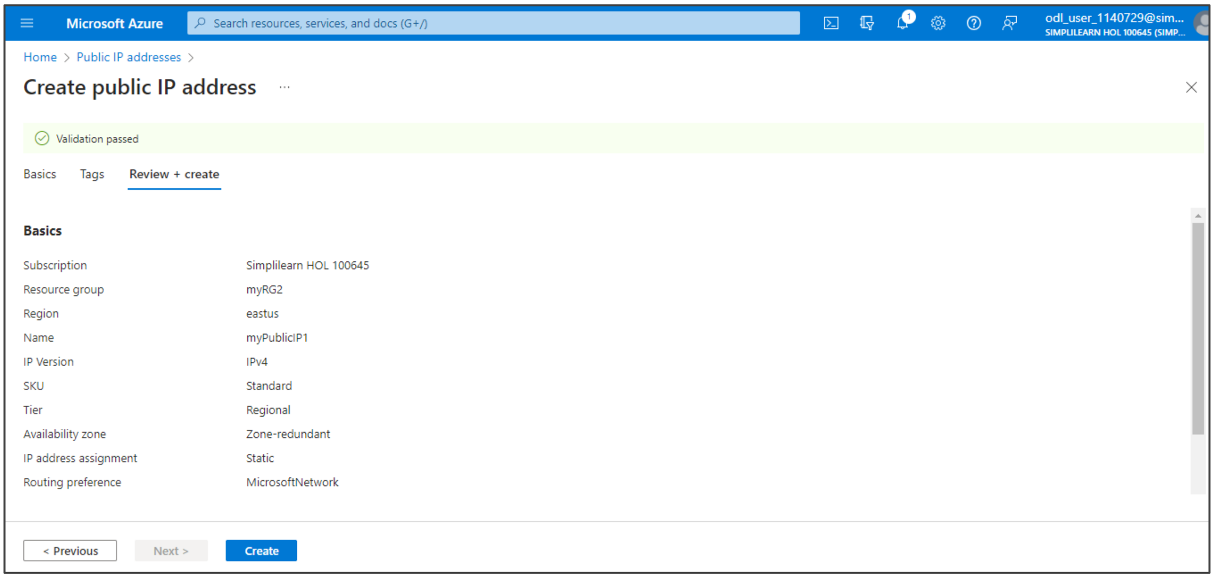




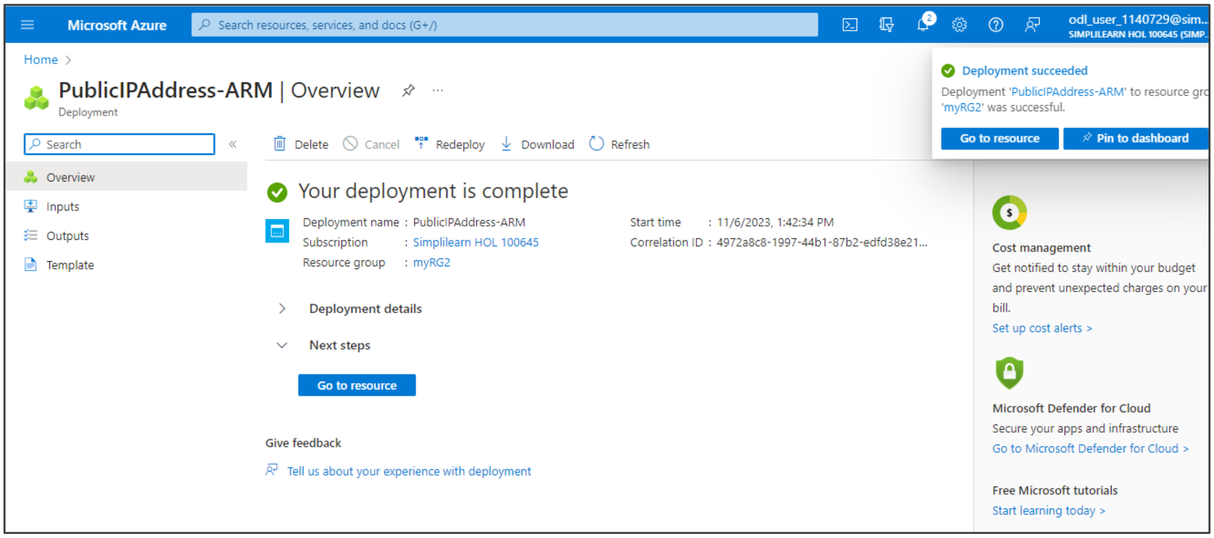
3.5 Click on **Review + create**



3.6 After the validation passes, click on **Create**

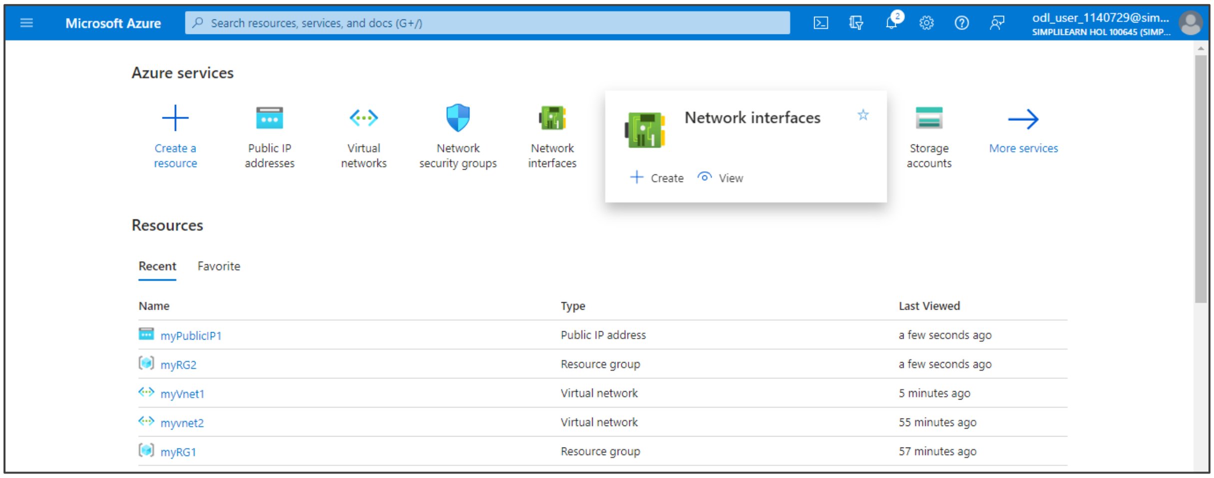


The public IP address is created as shown below:

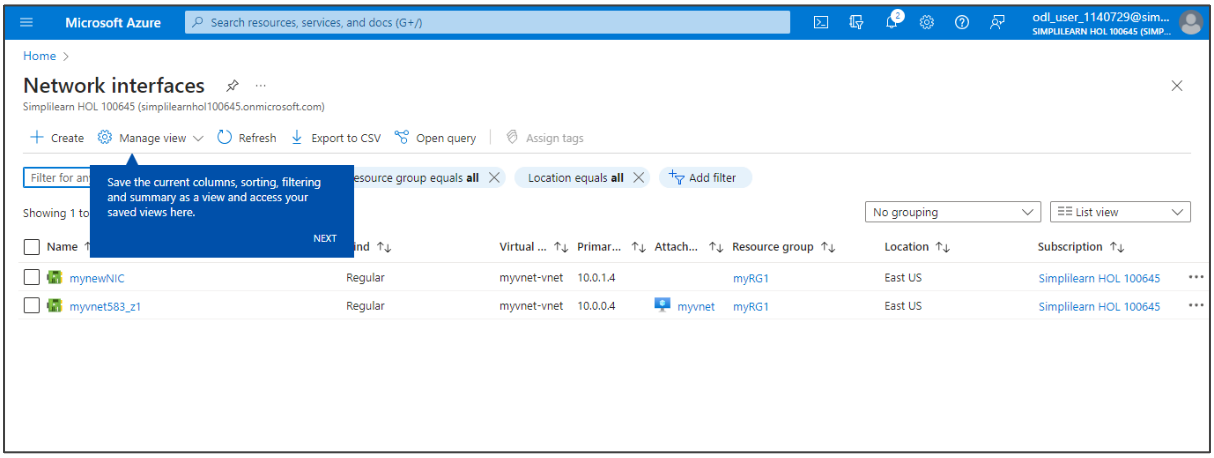


Step 4: Create a network interface

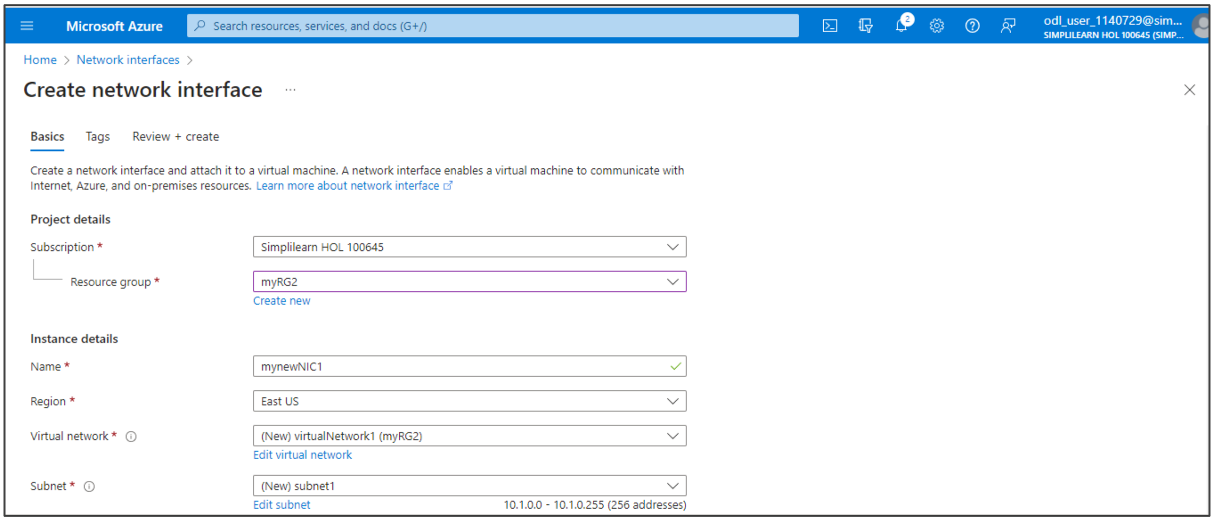
4.1 In the Azure portal, search for and select **Network interfaces**

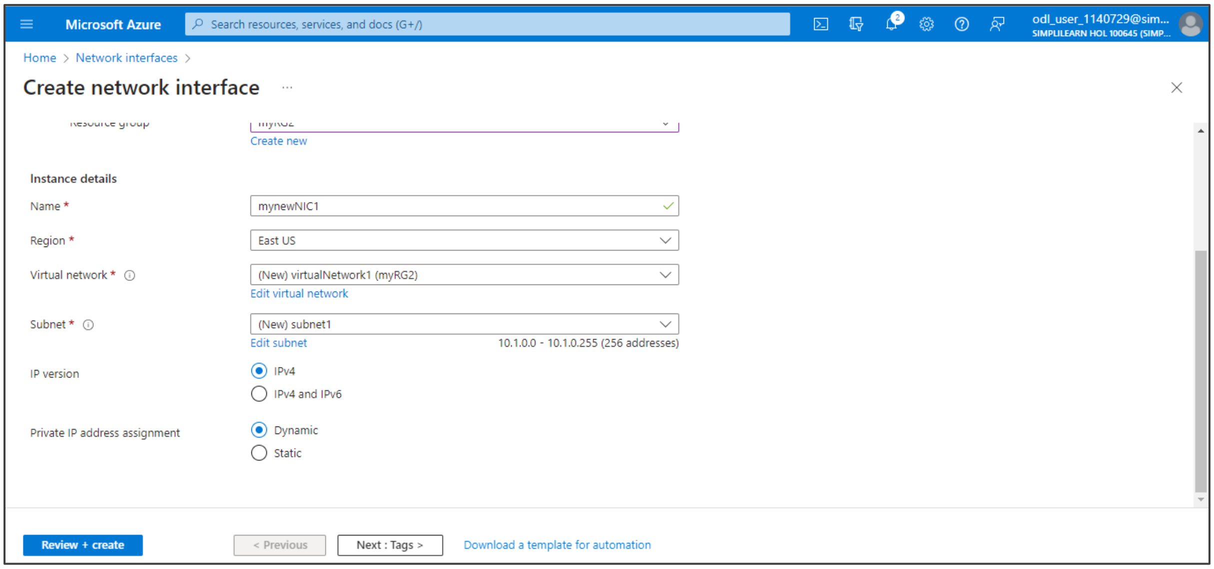


4.2 In the **Network interfaces** page, click on **+ Create**

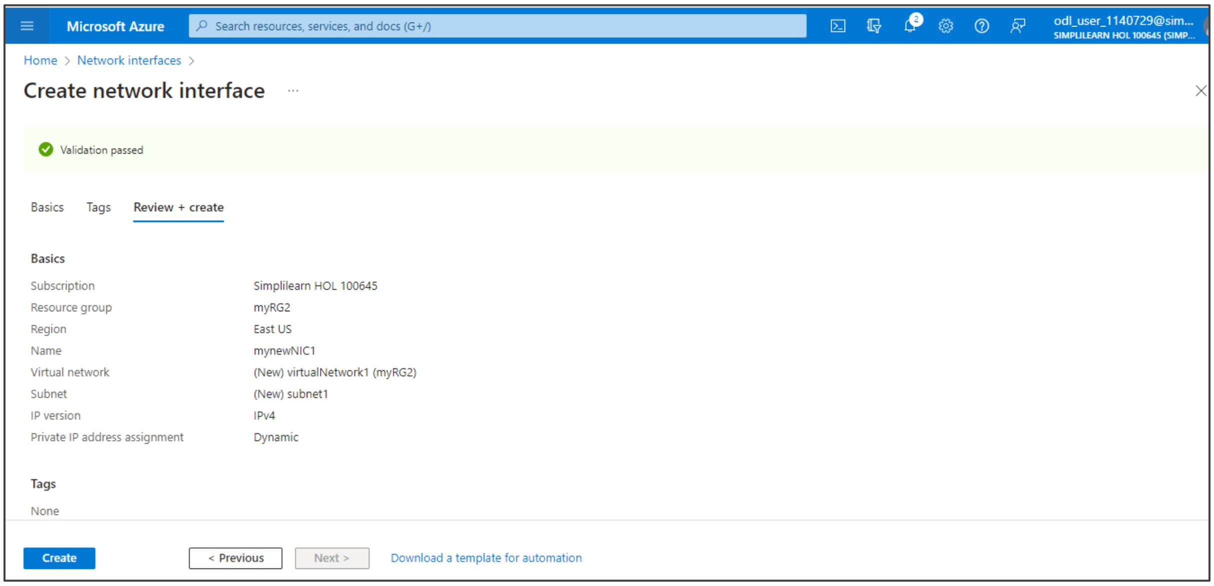


4.3 In the **Create network interface** page, navigate to the **Basics** tab and enter the following details as shown below. Then, click on **Review + create.**

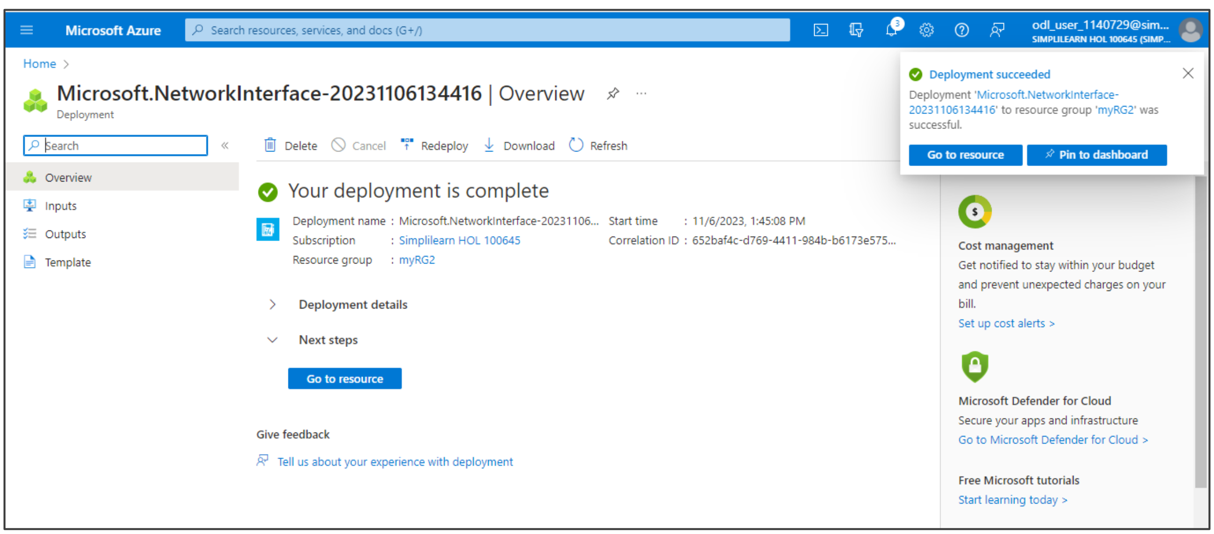




4.4 After the validation passes, click on **+ Create**

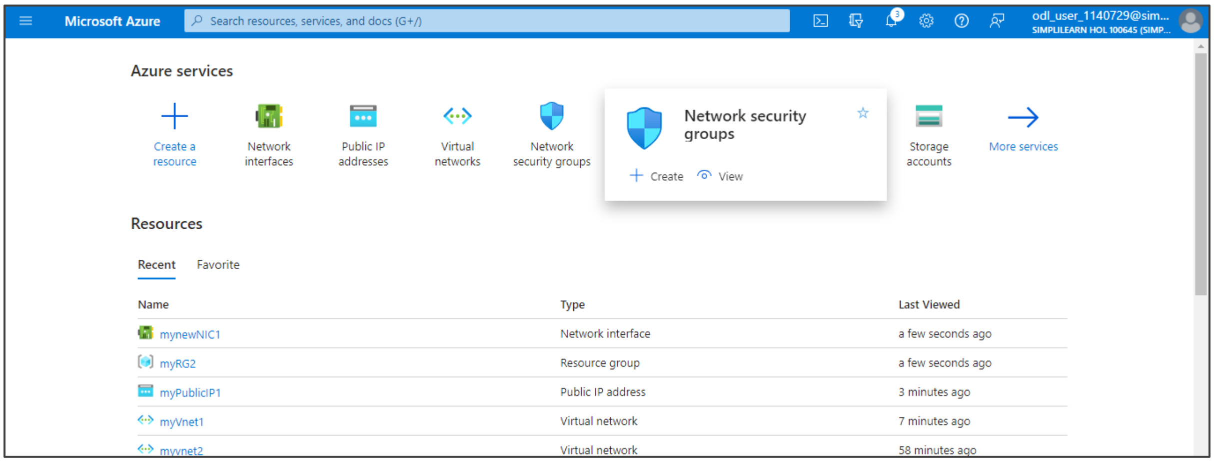


The network interface is created as shown below:

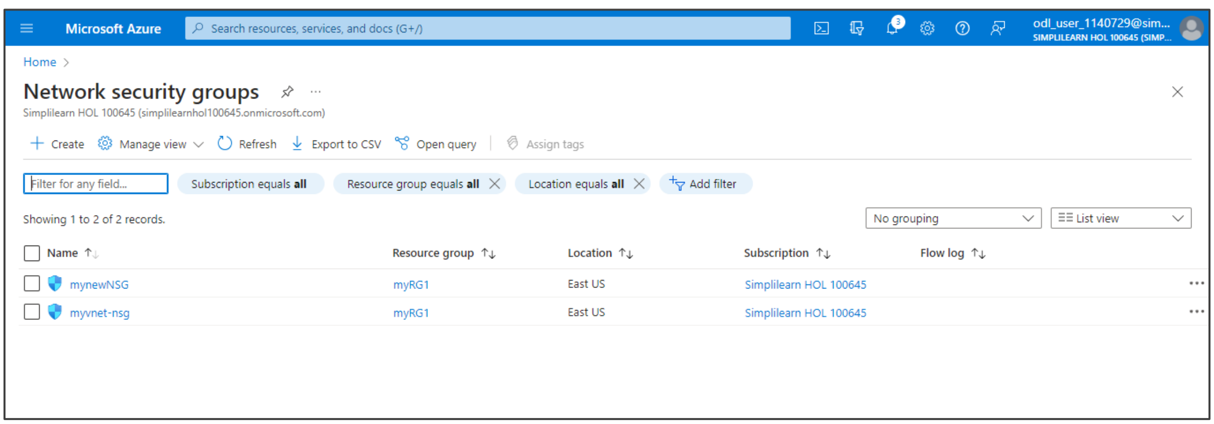


Step 5: Create a network security group

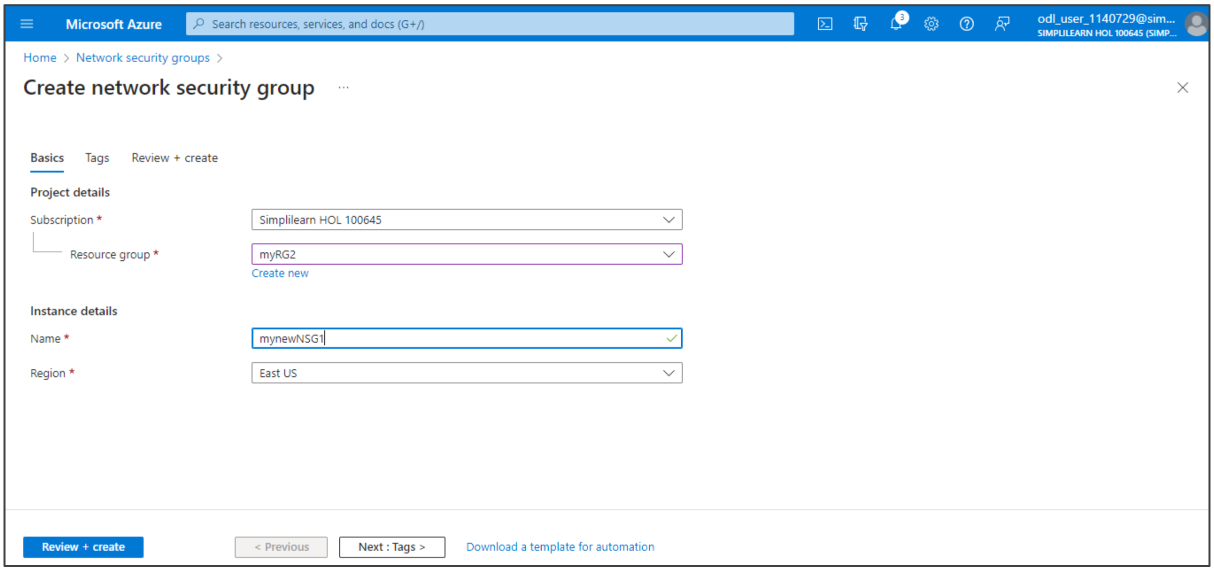
5.1 In the Azure portal, search for and select **Network security groups**



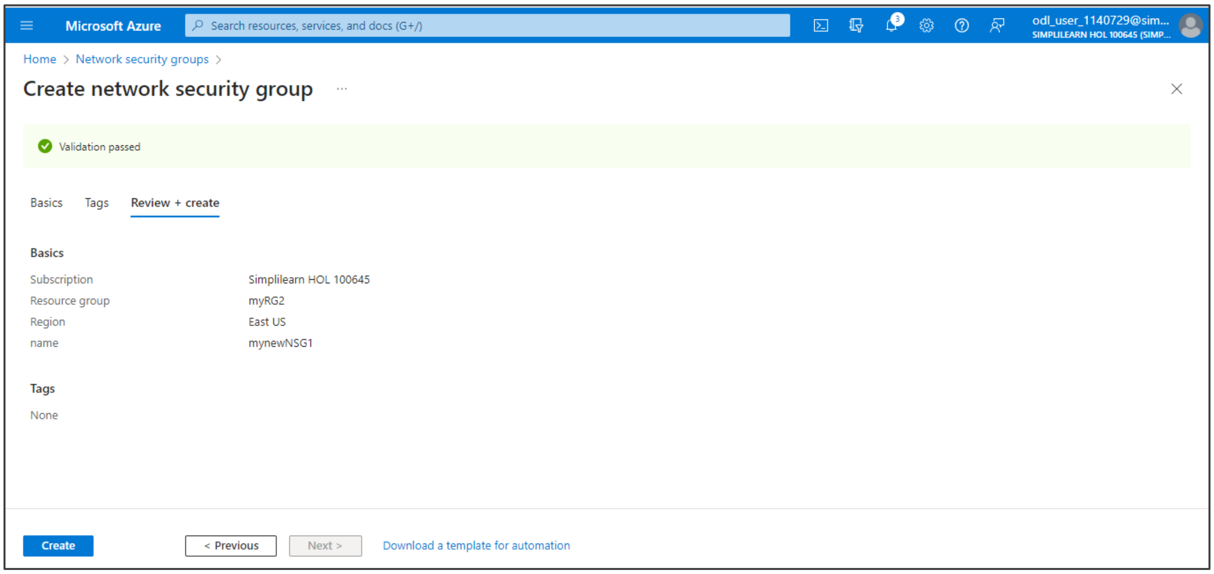
5.2 In the **Network security groups** page, click on **+ Create**



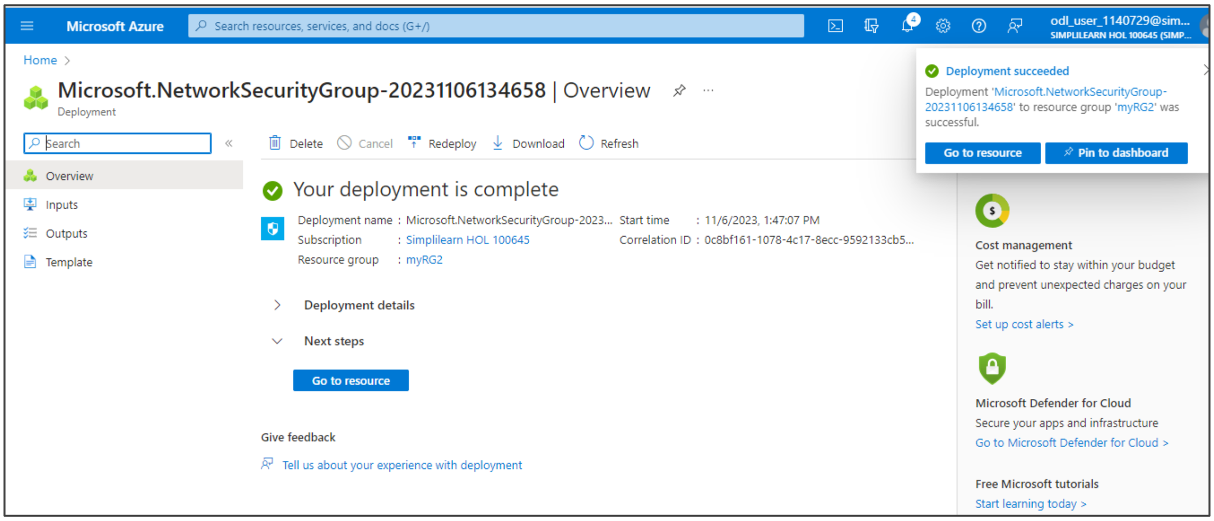
5.3 In the **Create network security group** page, navigate to the **Basics** tab and enter the details as shown below. Then, click on **Review + create.**



5.4 After the validation passes, click on **Create**



The network security group is created as shown below:



By following these steps, you have successfully created Azure's virtual networking capabilities by creating a virtual network for Azure virtual machines, implementing network-based segmentation with static IP addresses, securing public endpoints, and enabling DNS name resolution within the virtual network.